

How the PetHaven system works

PetHaven inventory prototype — a plain-language guide to what happens when customers buy in a shop or online

The big picture

PetHaven has **five shops** in Sydney and an **online store**. Three separate computer systems run the business:

1. **The store system** — the tills and stock records in the five shops. It always knows exactly what's on each shelf.
2. **The delivery system** — records when suppliers drop off new stock.
3. **The online store** — the website. It shows customers **one number per product**: how many are available across all five shops combined.

The catch is that the website's number isn't live. It only gets corrected when someone runs a **sync**. Between syncs it can be out of date ("stale").

Behind all three systems sits a **data warehouse**: one central record of every stock movement in the business. The website's number is calculated from it, and all the reports read from it.

Each system also uses its own codes. For example, the store system calls the dog food *9300601001019*, the delivery system calls it *PF-DOG-ADT-3K* and the website calls it *WEB-10001*. The warehouse translates all of these into one code, *P001*, so everything lines up.

1. A customer buys in a shop

Example: a customer at Parramatta buys 2 bags of dog food at the till.

1. **The shop's stock drops straight away.** Parramatta had 37 bags; now it has 35.
2. **The sale is copied to the warehouse automatically**, in the same instant:
 - **Copy:** the sale is copied exactly as the till recorded it, using the store's own codes.
 - **Translate:** the store number and barcode are converted into the warehouse's codes (Parramatta becomes S01, the dog food becomes P001).
 - **Check:** if a code isn't recognised (for example a brand-new product nobody has registered yet), the record is set aside with a note explaining why, instead of being guessed. It loads as soon as someone approves the new code.
 - **Save:** the warehouse records "Parramatta, dog food, -2, at this time".
3. **The website doesn't change.** It still says 183 available, even though there are really 181. This is the out-of-date number the project is about, and it stays that way until the next sync.

If the customer buys several things at once, they all go on one receipt. If even one item isn't on the shelf, the whole sale is refused.

2. A customer buys online

Example: a customer in Bondi buys 2 bags of dog food online.

1. **They add items to their bag.** The website only allows what its number says is in stock.
2. **They press checkout. Before any money is taken,** the system checks the **real** stock in the shops — first the shop **closest to the customer** (here, Bondi Junction), which is where they'll pick up the order. Bondi has enough, so it's chosen.
3. **Payment goes through and the order is created.** Bondi sets the 2 bags aside behind the counter: 2 fewer on the shelf, 2 more "reserved".
4. **The website number drops straight away** (183 → 181). The website knows about its own sales.
5. **The warehouse records the reservation** automatically, as with shop sales.
6. **The customer collects** at Bondi, and the reserved bags leave the shop.

If the customer cancels before collecting, the items go back on the shelf. The website only finds out at the next sync.

3. An online customer buys items that are in different shops

Example: a Bondi customer buys a squeaky duck and an aquarium kit.

At checkout the system looks for each item separately:

- **Squeaky duck:** Bondi has it, so it's set aside at Bondi.
- **Aquarium kit:** Bondi has none. The system tries the next-closest shops and finds one at **Newtown**, which sets it aside.

Payment goes through, and the customer still collects **everything at Bondi**. The Newtown kit has to be **transferred**:

1. **Sent:** Newtown sends the kit. While it's on the way, it doesn't count as stock in any shop.
2. **Received:** Bondi receives it and sets it aside with the duck.
3. **Collected:** only now can the customer collect, because the order is ready only when **every** item has arrived. The system won't let them collect half an order, and the order can't be cancelled while something is on the way.

Every step (set aside, sent, received, collected) is recorded in the warehouse, so the reports always show where each item is: *waiting to be sent, in transit or ready for collection*.

4. The website shows stock that isn't really there

This is the core business problem.

Example from the sample data: the website was synced when there were 7 aquarium kits. Since then, 5 were sold in the shops. The website still says 7, but only 2 are really left — one at Newtown and one at Penrith.

1. A Bondi customer puts **3 kits** in their bag. The website says 7, so it lets them.
2. They press checkout. **Before payment**, the system checks the real shop stock and finds that no shop has 3 kits. There are only 2 in total.

3. **Checkout is blocked.** The customer isn't charged, nothing is set aside, and they're shown which item can't be supplied. Their bag stays open, so they can remove it and buy the rest.
4. The warehouse records this as a **blocked checkout**, and **Report 3** lists it: *"website showed 7, really 2: stock sold since last sync, online number was stale."*

A checkout can also be blocked when there's enough in total but **no single shop** has the full amount — for example, 2 dog beds when the shops have 1 each. Report 3 gives that reason separately: *"enough stock in total but no single store had enough."*

Why this matters: nobody is ever charged for something that can't be supplied. But the customer was still shown stock that didn't exist and walked away disappointed: a lost sale. That's the cost of the out-of-date number, and the reports measure it.

5. The sync puts the website right

Someone runs the sync whenever they want; it isn't automatic. When they do, it:

1. Takes every stock movement since the last sync (shop sales, deliveries, online orders, cancellations).
2. Recalculates each shop's real totals and the correct website number for every product.
3. Updates the website.
4. Records a **before and after** for every number that changed — for example, *"aquarium kit: website 7 → 2"*.
5. Checks that the warehouse agrees with every shop's own records.

Right after a sync the website is correct again, until the next shop sale or delivery makes it stale.

In one line each

Situation	What happens
Shop sale	Shelf drops instantly, warehouse updated instantly, website unchanged until sync
Delivery	Shelf rises instantly, warehouse updated instantly, website unchanged until sync
Online order, closest shop has everything	Checked before payment, set aside there, website drops instantly, customer collects
Online order, items in different shops	Each item set aside where it's found, sent to the closest shop, collected together once all have arrived
Website shows stock that's gone	Checkout blocked before payment, no charge, recorded as a lost sale in Report 3
Sync	Website corrected from the warehouse, every change logged before → after